

Quiz 1

Name:

- 1) For a forward-biased diode, the barrier potential _____ as temperature increases.
 a) Decreases b) remains constant c) Increases
- 2) An n-type semiconductor material
 a) is intrinsic. c) has trivalent impurity atoms added.
 b) has pentavalent impurity atoms added. d) requires no doping.
- 3) For a forward-biased diode, as temperature is _____, the forward current _____ for a given value of forward voltage.
 a) decreased, increases c) increased, increases
 b) increased, decreases d) decreased, decreases
- 4) A silicon diode measures a low value of resistance with the meter leads in both positions. The trouble, if any, is
 a) the diode is open. c) the diode is shorted to ground.
 b) the diode is internally shorted. d) the diode is working correctly.
- 5) Single-element semiconductors are characterized by atoms with _____ valence electrons.
 a) 3 b) 4 c) 5 d) 2 e) none of the above
- 6) Under normal conditions a diode conducts current when it is
 a) Reverse-biased c) forward-biased
 b) Avalanched d) saturated
- 7) A diode conducts when it is forward-biased, and the anode is connected to the _____ through a limiting resistor.
 a) Positive supply c) negative supply
 b) Cathode d) anode
- 8) As the forward current through a silicon diode increases, the internal resistance
 a) Increases b) decreases c) remains the same
- 9) The boundary between p-type material and n-type material is called
 a) A diode c) a reverse bias diode
 b) A pn junction d) a forward biased diode
- 10) An ideal diode presents a(n) _____ when reversed-biased and a(n) _____ when forward-biased.
 a) Open, short c) short, open
 b) Open, open d) short, short
- 11) Current carriers are : 1) 2)
- 12) The symbol of diode :
- 13) a) V_D for silicon :
 b) V_D for germanium :
- 14) In reverse bias condition, the positive side of battery will connect to the of the diode.
- 15) On the graph, suppose Ge diode: complete it:

